Heaps

Chapter 17

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The ADT Heap

- A heap is a complete binary tree that either is
  - Empty or …
  - Whose root contains a value ≥ each of its children and has heaps as its subtrees
- It is a special binary tree … different in that
  - It is ordered in a weaker sense
  - it will always be a complete binary tree

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The ADT Heap

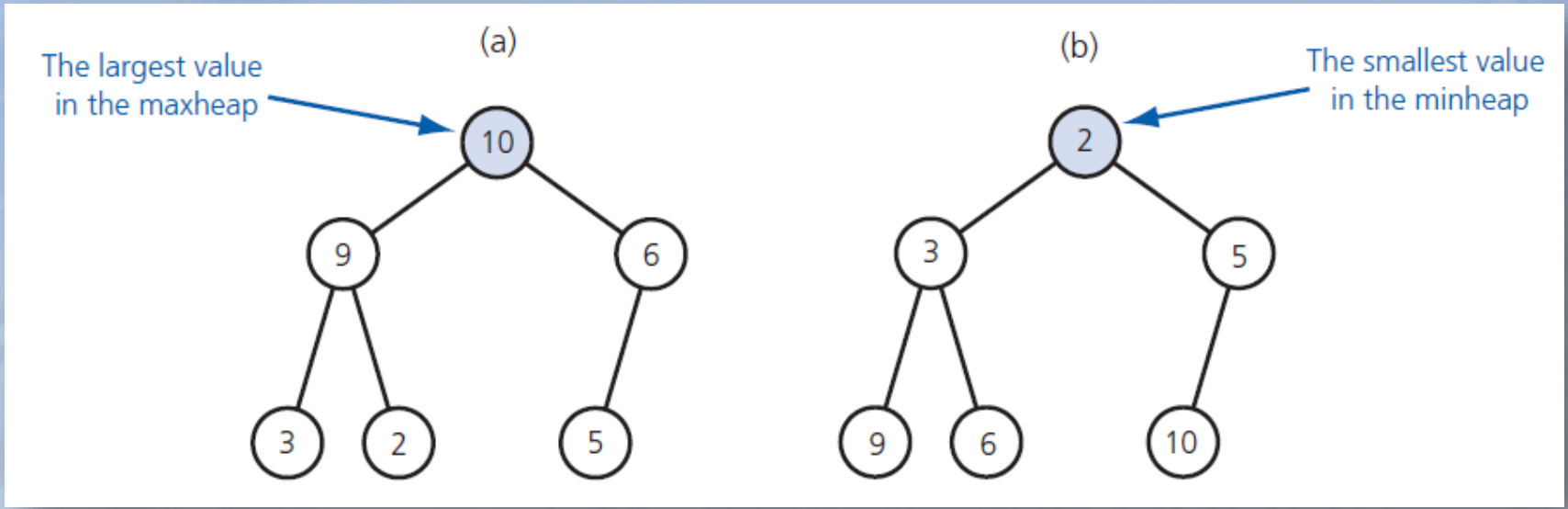

FIGURE 17-1 (a) A maxheap and (b) a minheap

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The ADT Heap

```
Heap
---
---
+isEmpty(): boolean
+getNumberOfNodes(): integer
+getHeight(): integer
+peekTop(): ItemType
+add(newData: ItemType): boolean
+remove(): boolean
+clear(): void
```

FIGURE 17-2 UML diagram for the class Heap

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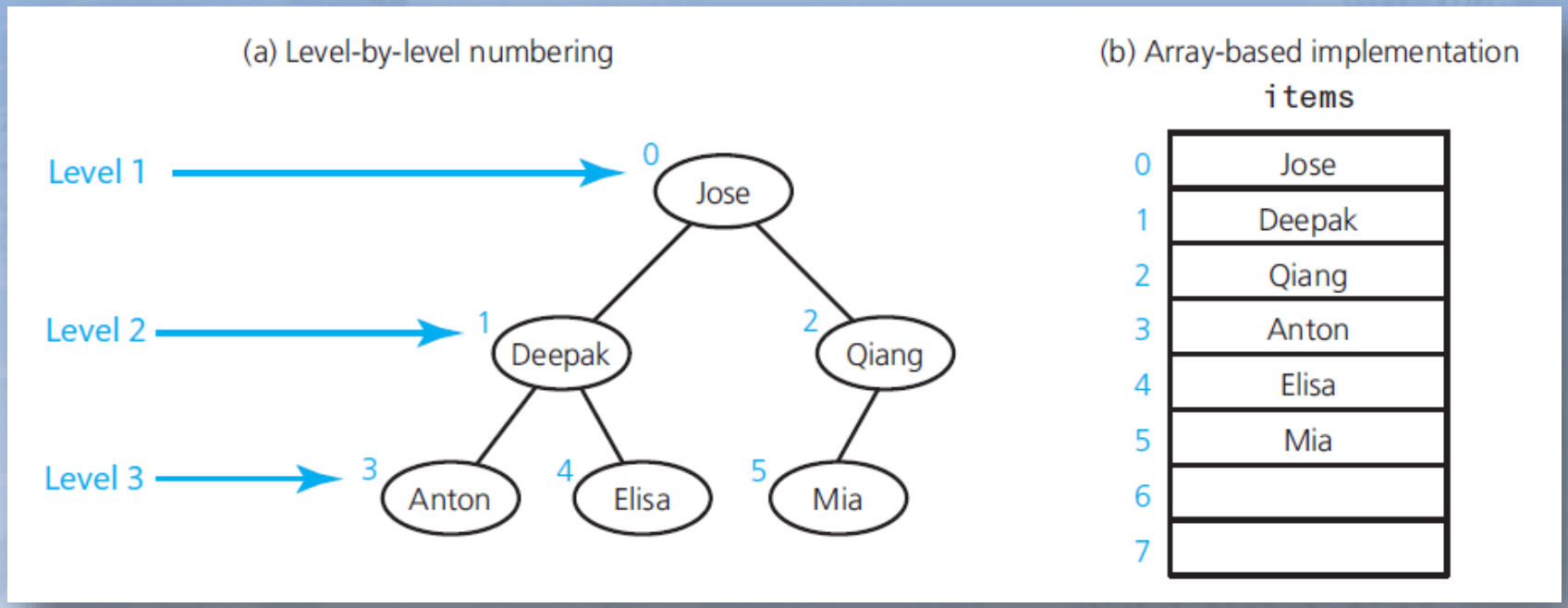
Array-Based Implementation of a Heap



Given any node items[i], you can easily locate both of its children and its parent:

- Its left child, if it exists, is items[2 * i + 1]
- Its right child, if it exists, is items[2 * i + 2]
- Its parent, if it exists, is items[(i - 1) / 2]

Figure 17-3 A complete binary tree and its array-based implementation

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Algorithms for Array-Based Heap Operations

- Assume following private data members
  - items: an array of heap items
  - itemCount: an integer equal to the number of items in the heap
  - maxItems: an integer equal to the maximum capacity of the heap

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Algorithms for Array-Based Heap Operations

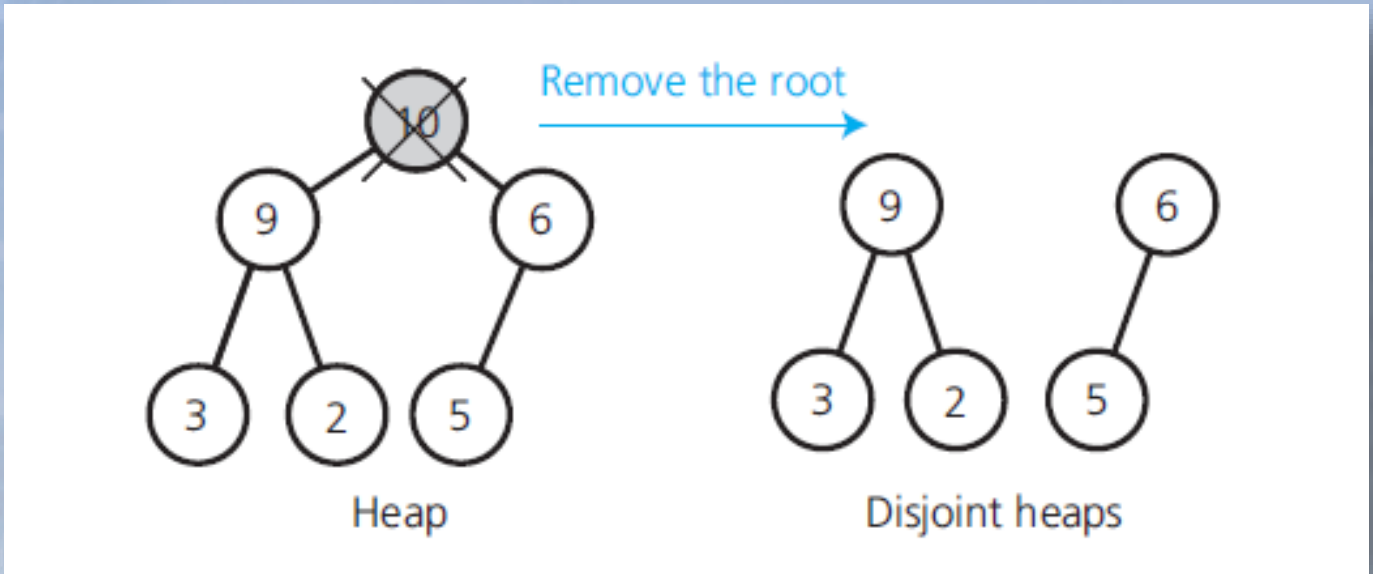

FIGURE 17-4 Disjoint heaps after removing the heap's root

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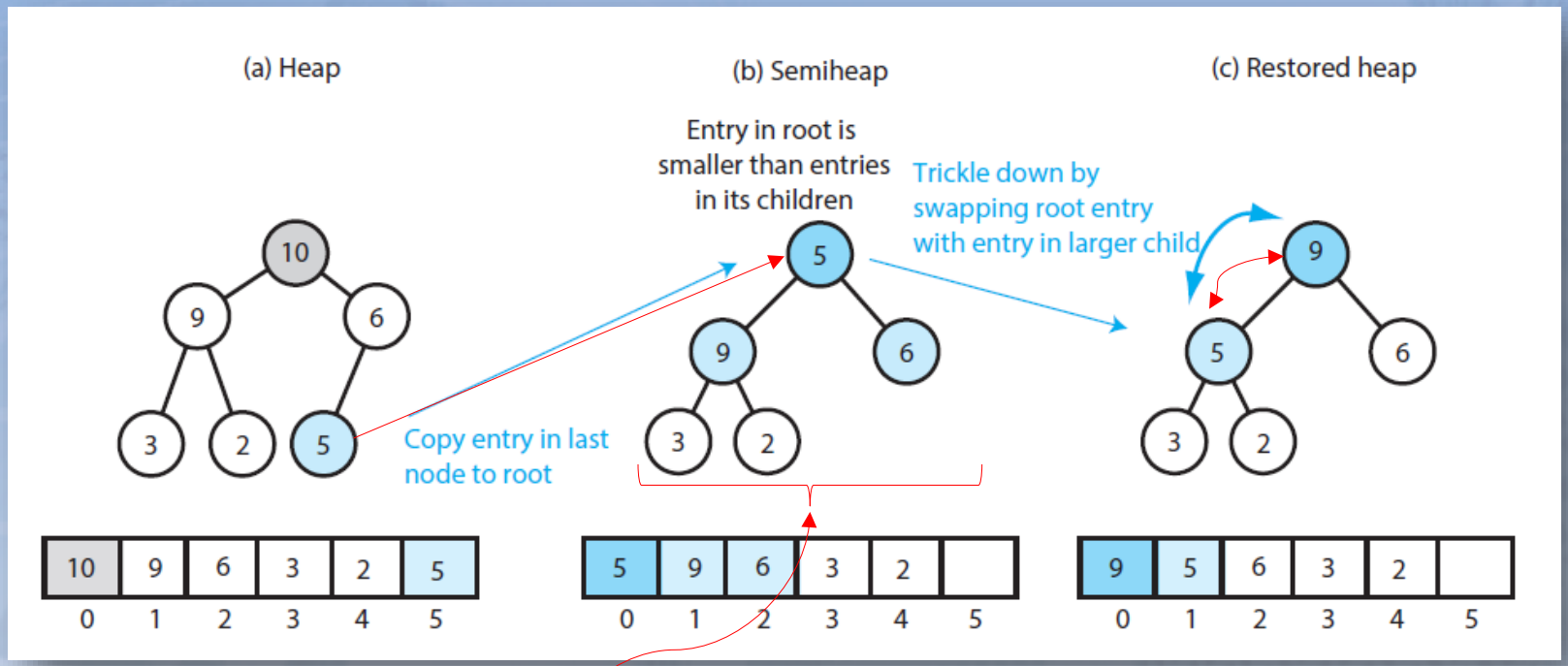
Algorithms for Array-Based Heap Operations



Remove root item (10) from a heap.

1. remove the last node of the tree and place its item in the root
2. If it is not a heap, then it is a semiheap. Recursively, trickle down the item in the root until it reaches a node in which it will not be out of place (heapify)

Removal efficiency.

- This item travels down a single path from the root to, at worst, a leaf.
- The number of array items it must swap is no greater than the height of the tree.
- Thus, remove is O(log n), which is in fact quite efficient.

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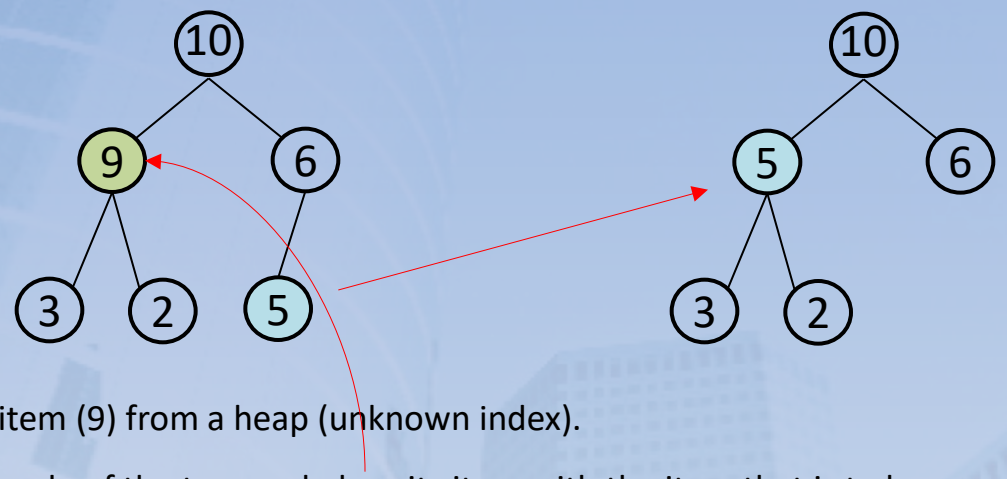
Algorithms for Array-Based Heap Operations



Remove an arbitrary item (9) from a heap (unknown index).

1. remove the last node of the tree and place its item with the item that is to be removed
2. If it is not a heap, then it is a semiheap. Recursively, trickle down the item in the root until it reaches a node in which it will not be out of place (heapify)

Removal efficiency.

- Find the arbitrary node which is O(n)
- Swap with the last node which is O(1)
- Remove the last node which is O(1)
- Heapify (up or down) which is O(log n)
- Thus, to remove an arbitrary node is O(n + log n), which is O(n).

Algorithms for Array-Based Heap Operations

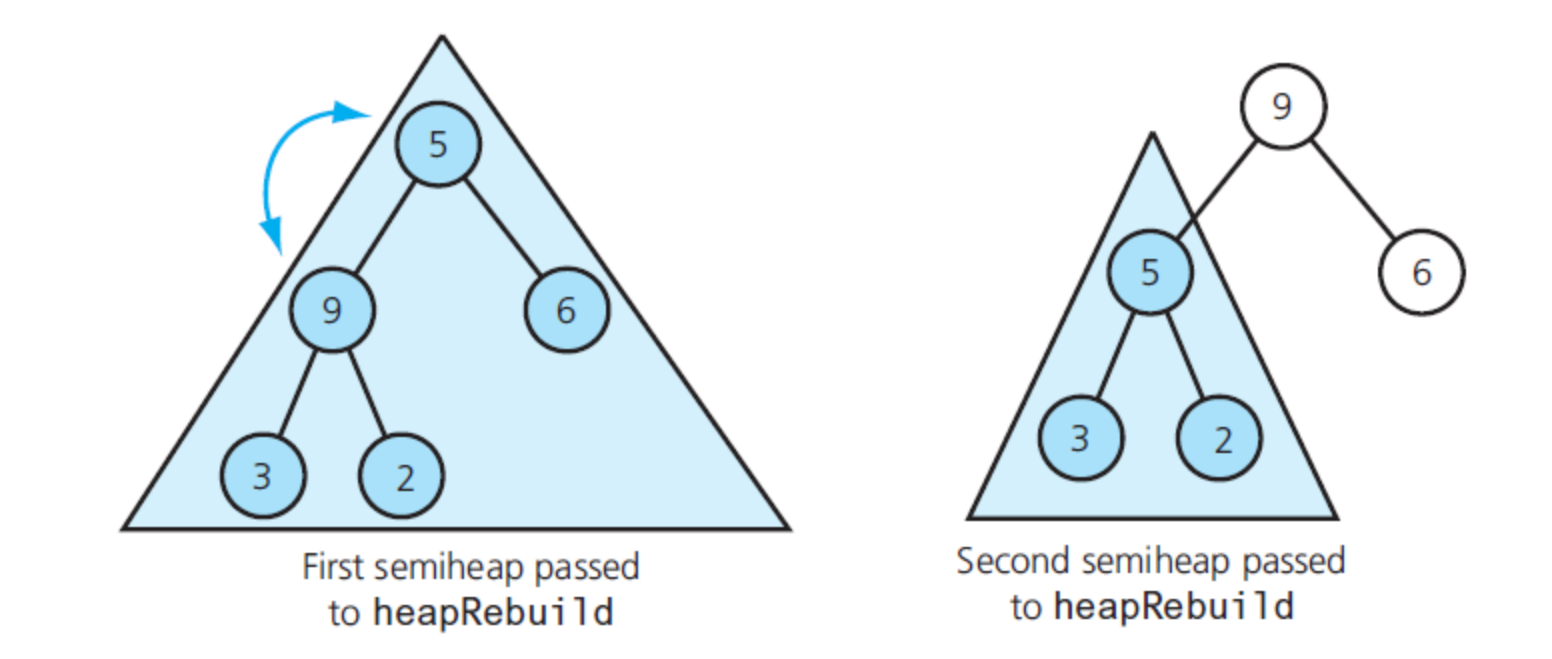


FIGURE 17-6 Recursive calls to heapRebuild

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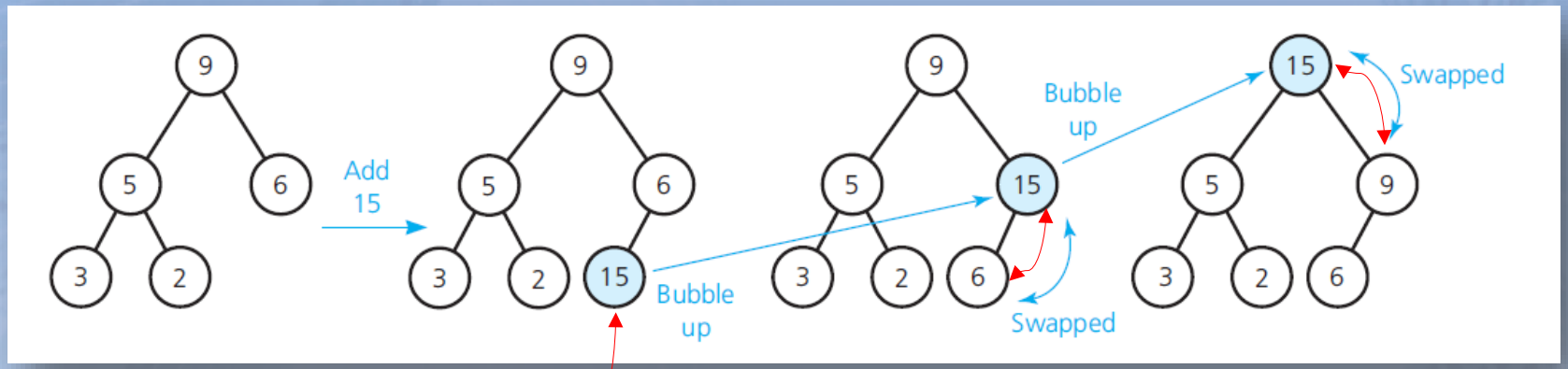
Algorithms for Array-Based Heap Operations



Add an item (15) to a heap (opposite of remove).

1. Insert new item at the bottom of the tree
2. Recursively trickle up to its proper place (the parent of the node in items[i] is always stored in items[(i - 1) / 2] —unless, of course, the node is the root).

Add efficiency.

- This item travels down a single path from a leaf to the root, at worst.
- The number of array items it must swap is no greater than the height of the tree.
- Thus, add is O(log n), which is in fact quite efficient.

FIGURE 17-5 Adding 15 to a heap

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Transforming Binary Tree to Heap

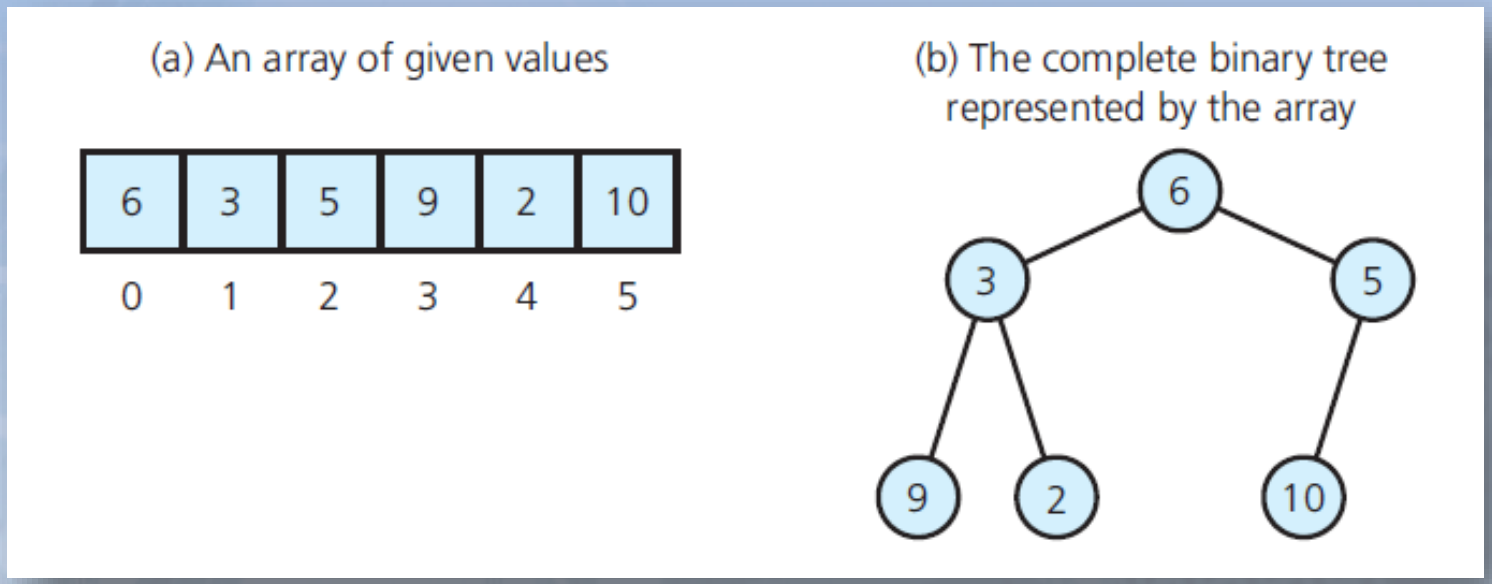

FIGURE 17-8 Array and its corresponding complete binary tree

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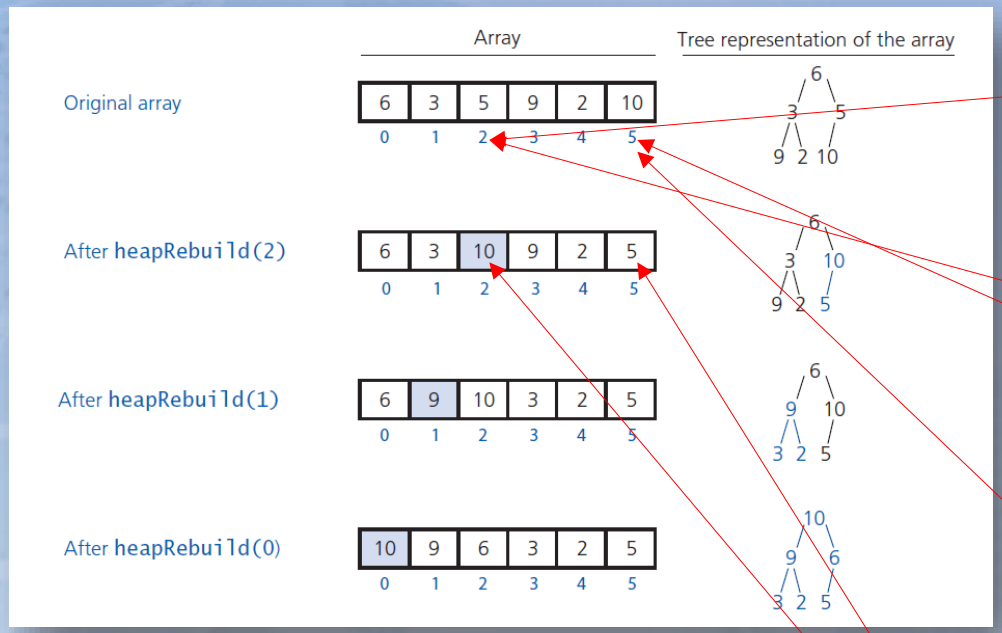
Transforming Binary Tree to Heap




```
function buildMaxHeap(array):
    n = length of array

    # Step 1: Start from the last non-leaf node and move upwards
    for i from n/2 - 1 down to 0:  # from index 2, 1, 0
        heapRebuild(array, n, i)

function heapRebuild(array, n, i):
    largest = i    # initialize largest to root
    leftChild = 2 * i + 1  # index 5, array[leftChild] = 10
    rightChild = 2 * i + 2  # none

    # Step 2: Check if the left child exists and is greater than the
current largest
    if leftChild < n and array[leftChild] > array[largest]:
        largest = leftChild

    # Step 3: Check if the right child exists and is greater than the
current largest
    if rightChild < n and array[rightChild] > array[largest]:
        largest = rightChild

    # Step 4: If largest is not the current node, swap and
heapRebuild the affected subtree
    if largest != i:
        swap(array[i], array[largest])
        heapRebuild(array, n, largest)
```

FIGURE 17-9 Transforming an array into a heap

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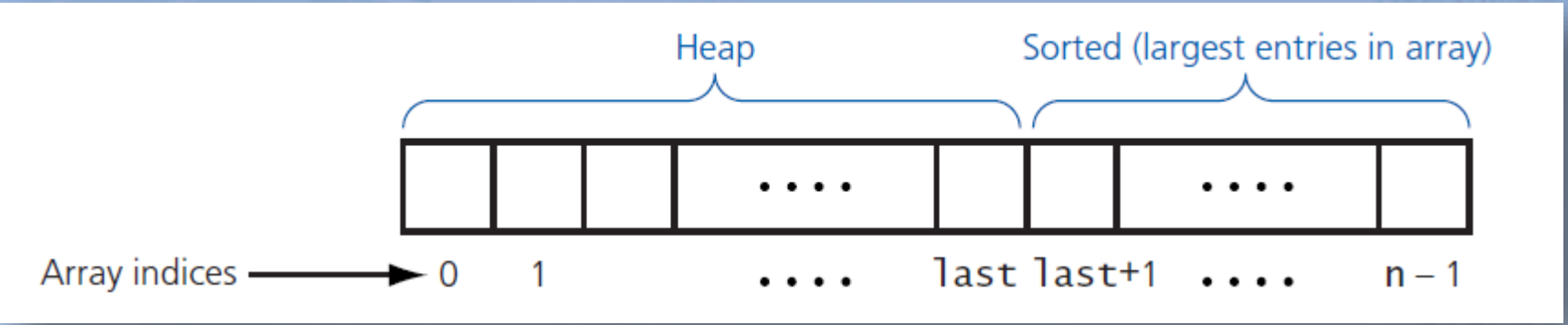
Heap Sort



heap sort algorithm

1. Transform array into heap
2. For each item from heap
   a. Swap the root (largest item) with last item in the heap
      Swap(myArray[0], myArray[last])
   b. Transform heap region to heap
   c. Decrement last /* reduce heap region/increase sorted region */

FIGURE 17-10 Heap sort partitions an array into two regions

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Heap Sort

heap sort algorithm

1. Transform array into heap
2. For each item from heap
   a. Swap the root (largest item) with last item in the heap
      Swap(myArray[0], myArray[last])
   b. Transform heap region to heap
   c. Decrement last /* reduce heap region/increase sorted region */

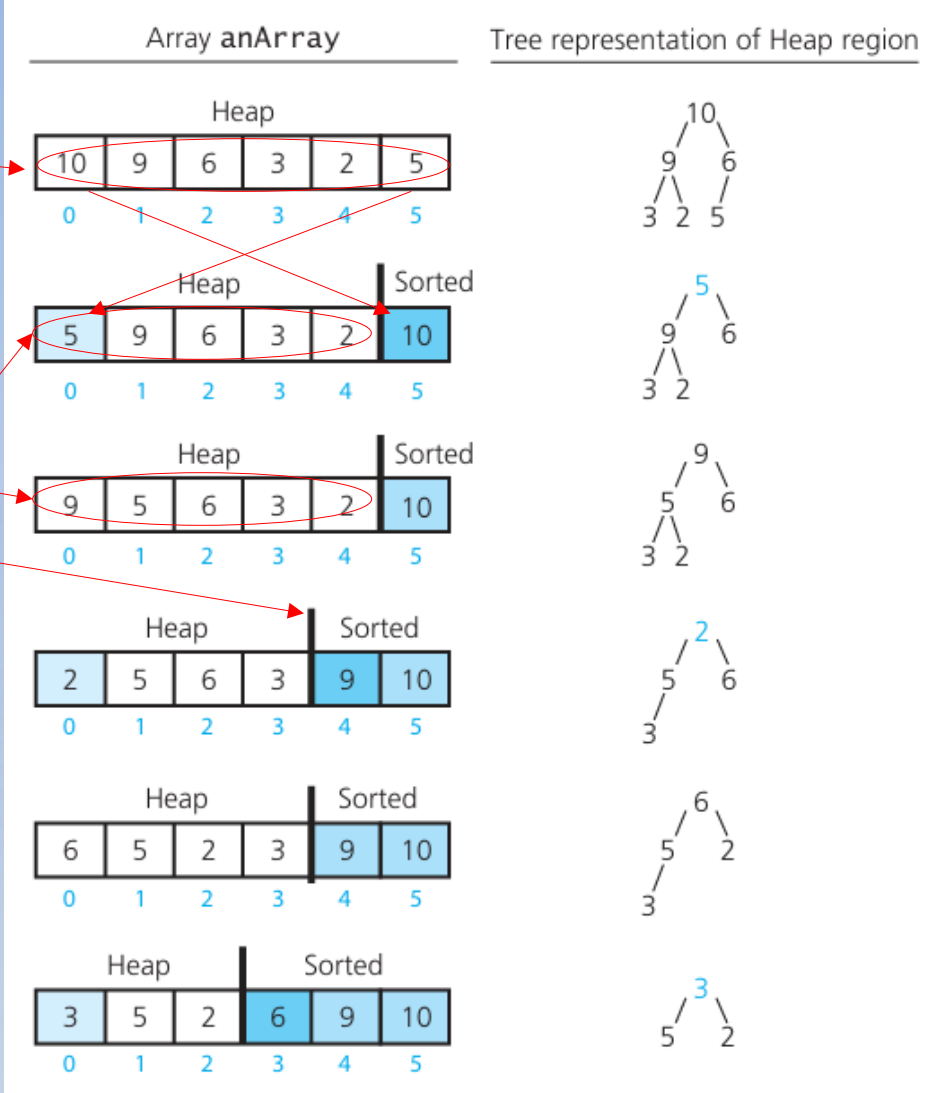

FIGURE 17-11 A trace of heap sort

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Heap Sort

heap sort algorithm

1. Transform array into heap
2. For each item from heap
   a. Swap the root (largest item) with last item in the heap
      Swap(myArray[0], myArray[last])
   b. Transform heap region to heap
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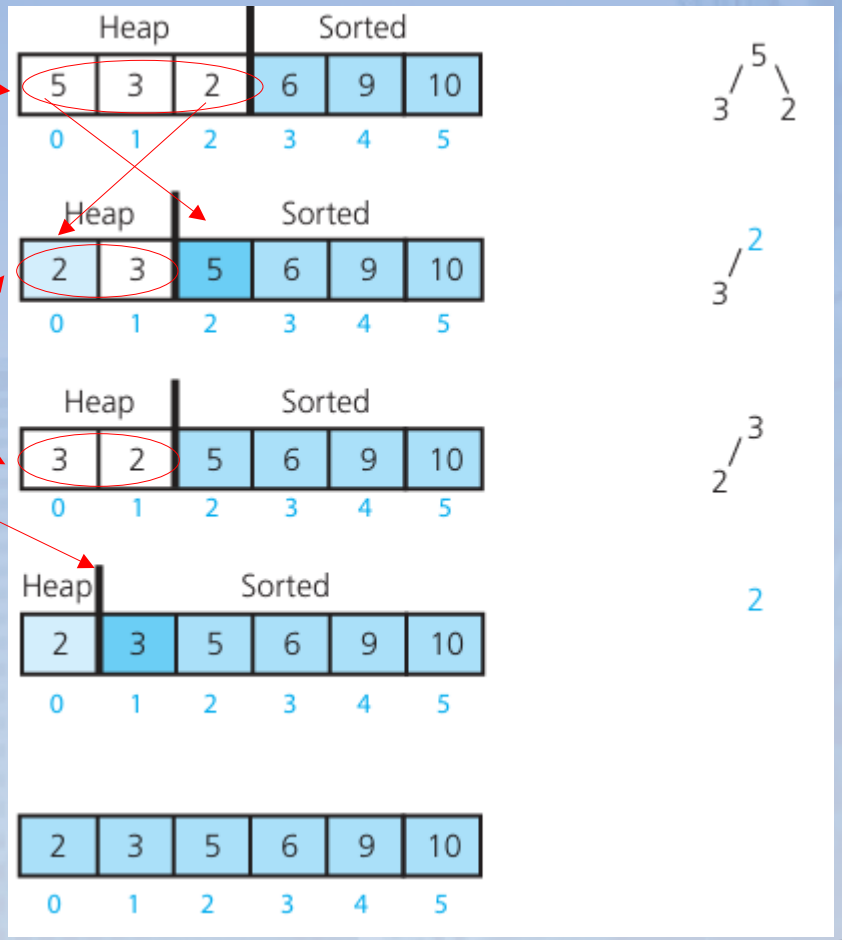

FIGURE 17-11 A trace of heap sort

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Heap Sort

- Heap sort converts an array into a heap to locate the array's largest item. This step enables the heap sort to sort an array in an efficient manner.
- Heap sort, like merge sort, has good worst-case and average-case behaviors
  - O(n * log n) // applies to the worst and average cases
- Heap sort has an advantage over merge sort in that it does not require a second array.

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End

Chapter 17

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Appendix

Chapter 17

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The ADT Heap

```
/** Interface for the ADT heap.
 @file HeapInterface.h */

#ifndef HEAP_INTERFACE_
#define HEAP_INTERFACE_

template<class ItemType>
class HeapInterface
{
public:
   /** Sees whether this heap is empty.
    @return  True if the heap is empty, or false if not. */
   virtual bool isEmpty() const = 0;

   /** Gets the number of nodes in this heap.
    @return  The number of nodes in the heap. */
   virtual int getNumberOfNodes() const = 0;
```

LISTING 17-1 An interface for the ADT heap

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The ADT Heap

```
    /** Gets the height of this heap.
     @return  The height of the heap. */
    virtual int getHeight() const = 0;

    /** Gets the data that is in the root (top) of this heap.
        For a maxheap, the data is the largest value in the heap;
        for a minheap, the data is the smallest value in the heap.
     @pre  The heap is not empty.
     @post  The root's data has been returned, and the heap is unchanged.
     @return  The data in the root of the heap. */
    virtual ItemType peekTop() const = 0;

    /** Adds a new data item to this heap.
```

LISTING 17-1 An interface for the ADT heap

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The ADT Heap

```
    /** Adds a new data item to this heap.
     @param newData  The data to be added.
     @post  The heap has a new node that contains newData.
     @return  True if the addition is successful, or false if not. */
    virtual bool add(const ItemType& newData) = 0;

    /** Removes the data that is in the root (top) of this heap.
     @return  True if the removal is successful, or false if not. */
    virtual bool remove() = 0;

    /** Removes all data from this heap. */
    virtual void clear() = 0;

    /** Destroys this heap and frees its assigned memory. */
     virtual ~HeapInterface() {  }
};  // end HeapInterface
#endif
```

LISTING 17-1 An interface for the ADT heap

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Algorithms for Array-Based Heap Operations

```
// Converts a semiheap rooted at index nodeIndex into a heap.
heapRebuild(nodeIndex: integer, items: ArrayType, itemCount: integer): void
{
   // Recursively trickle the item at index nodeIndex down to its proper position by
   // swapping it with its larger child, if the child is larger than the item.
   // If the item is at a leaf, nothing needs to be done.
   if (the root is not a leaf)
   {
      // The root must have a left child; find larger child
      leftChildIndex = 2 * rootIndex + 1
      rightChildIndex = leftChildIndex + 1
      largerChildIndex = rightChildIndex // Assume right child exists and is the larger
```

Recursive algorithm to transform semiheap to heap.

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Algorithms for Array-Based Heap Operations

```
      // Check whether right child exists; if so, is left child larger?
      // If no right child, left one is larger
      if ((largerChildIndex >= itemCount) ||
                                (items[leftChildIndex] > items[rightChildIndex]))
         largerChildIndex = leftChildIndex; // Assumption was wrong
      if (items[nodeIndex] < items[largerChildIndex])
      {
         Swap items[nodeIndex] and items[largerChildIndex]

         // Transform the semiheap rooted at largerChildIndex into a heap
         heapRebuild(largerChildIndex, items, itemCount)
      }
   }
   // Else root is a leaf, so you are done
}
```

Recursive algorithm to transform semiheap to heap.

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Algorithms for Array-Based Heap Operations

```
add(newData: itemType): boolean
{
   // Place newData at the bottom of the tree
   items[itemCount] = newData
   // Make new item bubble up to the appropriate spot in the tree
   newDataIndex = itemCount
   inPlace = false
   while ((newDataIndex >= 0) and !inPlace)
   {
      parentIndex = (newDataIndex - 1) / 2
      if (items[newDataIndex] <= items[parentIndex])
         inPlace = true
      else
      {
         Swap items[newDataIndex] and items[parentIndex]
         newDataIndex = parentIndex
      }
   }
   itemCount++
   return inPlace
}
```

Pseudocode for add

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The Implementation

```
/** Array-based implementation of the ADT heap.
 @file ArrayMaxHeap.h */
#ifndef ARRAY_MAX_HEAP_
#define ARRAY_MAX_HEAP_

#include "HeapInterface.h"
#include "PrecondViolatedExcept.h"

template<class ItemType>
class ArrayMaxHeap : public HeapInterface<ItemType>
{
private:
   static const int ROOT_INDEX = 0;          // Helps with readability
   static const int DEFAULT_CAPACITY = 21; // Small capacity for testing
   std::unique_ptr<ItemType[]> items;        // Array of heap items
   int itemCount;                            // Current count of heap items
   int maxItems;                             // Maximum capacity of the heap

```

LISTING 17-2 The header file for the class ArrayMaxHeap

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The Implementation

```
    // ------------------------------------------------------------------
    // Most of the private utility methods use an array index as a parameter
    // and in calculations. This should be safe, even though the array is an
    // implementation detail, since the methods are private.
    // ------------------------------------------------------------------

    // Returns the array index of the left child (if it exists).
    int getLeftChildIndex(const int nodeIndex) const;

    // Returns the array index of the right child (if it exists).
    int getRightChildIndex(int nodeIndex) const;

    // Returns the array index of the parent node.
    int getParentIndex(int nodeIndex) const;

    // Tests whether this node is a leaf.
    bool isLeaf(int nodeIndex) const;

```

LISTING 17-2 The header file for the class ArrayMaxHeap

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The Implementation

```
    // Converts a semiheap to a heap.
    void heapRebuild(int nodeIndex);

    // Creates a heap from an unordered array.
    void heapCreate();

public:
    ArrayMaxHeap();
    ArrayMaxHeap(const ItemType someArray[], const int arraySize);
    virtual ~ArrayMaxHeap();

    // HeapInterface Public Methods:
    bool isEmpty() const;
    int getNumberOfNodes() const;
    int getHeight() const;
    ItemType peekTop() const throw(PrecondViolatedExcept);
    bool add(const ItemType& newData);
    bool remove();
    void clear();
}; // end ArrayMaxHeap
#include "ArrayMaxHeap.cpp"
#endif
```

LISTING 17-2 The header file for the class ArrayMaxHeap

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The Implementation

```
template<class ItemType>
int ArrayMaxHeap<ItemType>::getLeftChildIndex(const int nodeIndex) const
{
   return (2 * nodeIndex) + 1;
}  // end getLeftChildIndex
```

Definition of method getLeftChildIndex

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The Implementation

```
template<class  ItemType>
ArrayMaxHeap<ItemType>::
ArrayMaxHeap(const ItemType someArray[], const int arraySize):
             itemCount(arraySize), maxItems(2 * arraySize)
{

    // Allocate the array
    items = std::make_unique<ItemType[]>(maxItems);

    // Copy given values into the array
    for (int i = 0; i < itemCount; i++)
       items[i] = someArray[i];

    // Reorganize the array into a heap
    heapCreate();
 }  // end constructor
```

Definition of the constructor

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The Implementation

```
for (index = itemCount / 2 down to 0)
{
   // Assertion: The tree rooted at index is a semiheap
   heapRebuild(index)
   // Assertion: The tree rooted at index is a heap
}
```

Building a heap from an array of data

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Transforming Binary Tree to Heap

```
template<class ItemType>
void ArrayMaxHeap<ItemType>::heapCreate()
{
   for (int index = itemCount / 2; index >= 0; index--)
      heapRebuild(index);
}  // end heapCreate
```

C++ method heapCreate

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The Implementation

```
template<class ItemType>
ItemType ArrayMaxHeap<ItemType>::peekTop() const throw(PrecondViolatedExcept)
{
   if (isEmpty())
      throw PrecondViolatedExcep("Attempted peek into an empty heap.");

   return items[0];
}  // end peekTop
```

C++ method peekTop which tests for an empty heap

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Heap Implementation of the ADT Priority Queue

```
/** ADT priority queue: Heap-based implementation.
 @file HeapPriorityQueue.h */
#ifndef HEAP_PRIORITY_QUEUE_
#define HEAP_PRIORITY_QUEUE_
#include "ArrayMaxHeap.h"
#include "PriorityQueueInterface.h"

template<class ItemType>
class HeapPriorityQueue : public PriorityQueueInterface<ItemType>,
                          private ArrayMaxHeap<ItemType>
{
public:
   HeapPriorityQueue();
   bool isEmpty() const;
   bool enqueue(const ItemType& newEntry);
   bool dequeue();

   /** @pre  The priority queue is not empty. */
   ItemType peekFront() const throw(PrecondViolatedExcept);
}; // end HeapPriorityQueue

#include "HeapPriorityQueue.cpp"
#endif
```

LISTING 17-3 A header file for the class HeapPriorityQueue

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Heap Implementation of the ADT Priority Queue

```
/** Heap-based implementation of the ADT priority queue.
 @file HeapPriorityQueue.cpp */

#include "HeapPriorityQueue.h"

template<class ItemType>
HeapPriorityQueue<ItemType>::HeapPriorityQueue()
{
   ArrayMaxHeap<ItemType>();
}  // end constructor

template<class ItemType>
bool HeapPriorityQueue<ItemType>::isEmpty() const
{
   return ArrayMaxHeap<ItemType>::isEmpty();
}  // end isEmpty

template<class ItemType>
bool HeapPriorityQueue<ItemType>::enqueue(const ItemType& newEntry)
{
   return ArrayMaxHeap<ItemType>::add(newEntry);
}  // end add
```

LISTING 17-4 An implementation of the class HeapPriorityQueue

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Heap Implementation of the ADT Priority Queue

```
template<class ItemType>
bool HeapPriorityQueue<ItemType>::dequeue()
{
   return ArrayMaxHeap<ItemType>::remove();
}  // end remove

template<class ItemType>
ItemType HeapPriorityQueue<ItemType>::peekFront() const throw(PrecondViolatedExcept)
{
   try
   {
      return ArrayMaxHeap<ItemType>::peekTop();
   }
   catch (PrecondViolatedExcept e)
   {
      throw PrecondViolatedExcept("Attempted peek into an empty priority queue.");
   }  // end try/catch
}  // end peekFront
```

LISTING 17-4 An implementation of the class HeapPriorityQueue

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Heap Implementation of the ADT Priority Queue

- Heap versus a binary search tree
  - If you know maximum number of items in the priority queue, heap is the better implementation.
- Finite, distinct priority values
  - Many items likely have same priority value
  - Place in same order as encountered

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Heap Sort

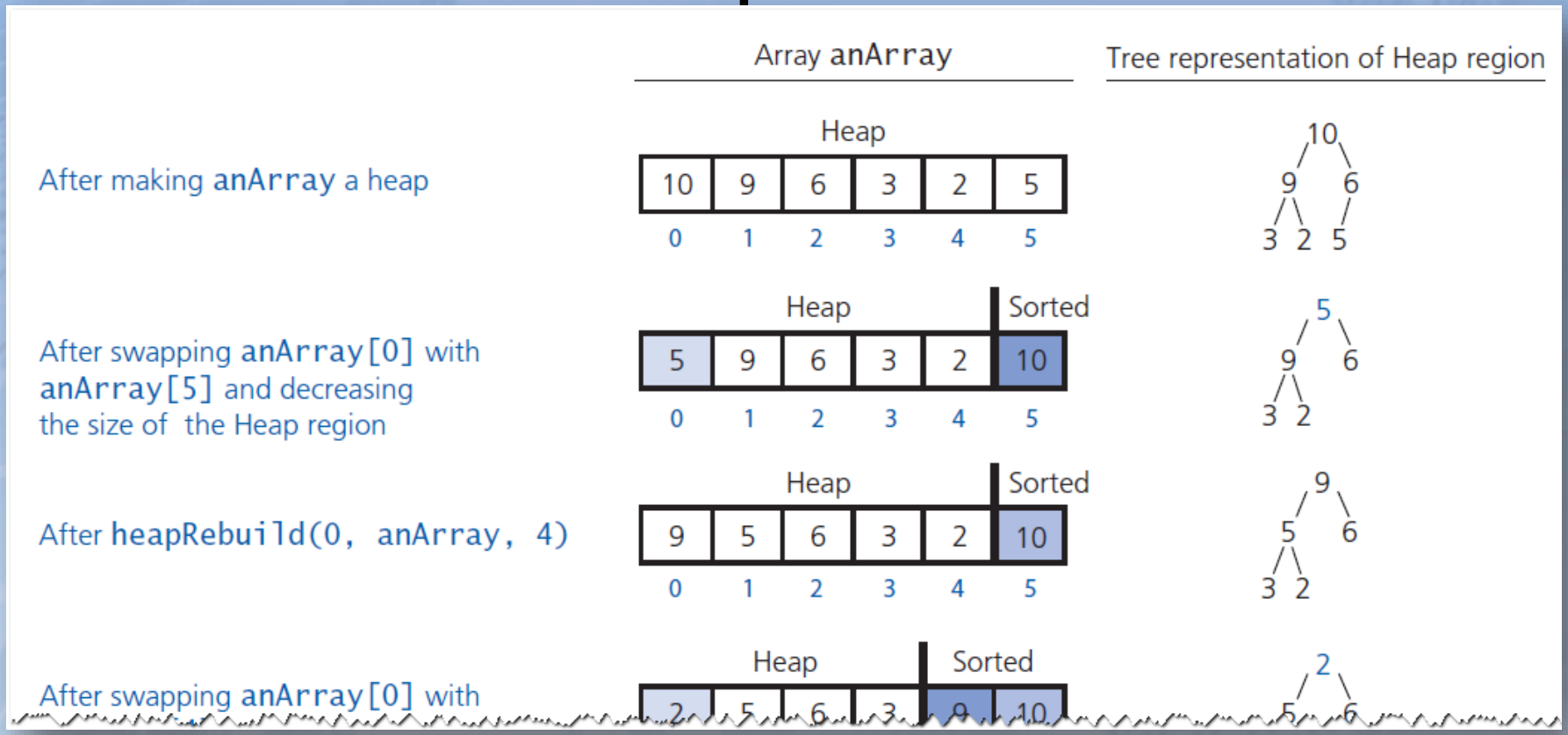

FIGURE 17-11 A trace of heap sort

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Heap Sort

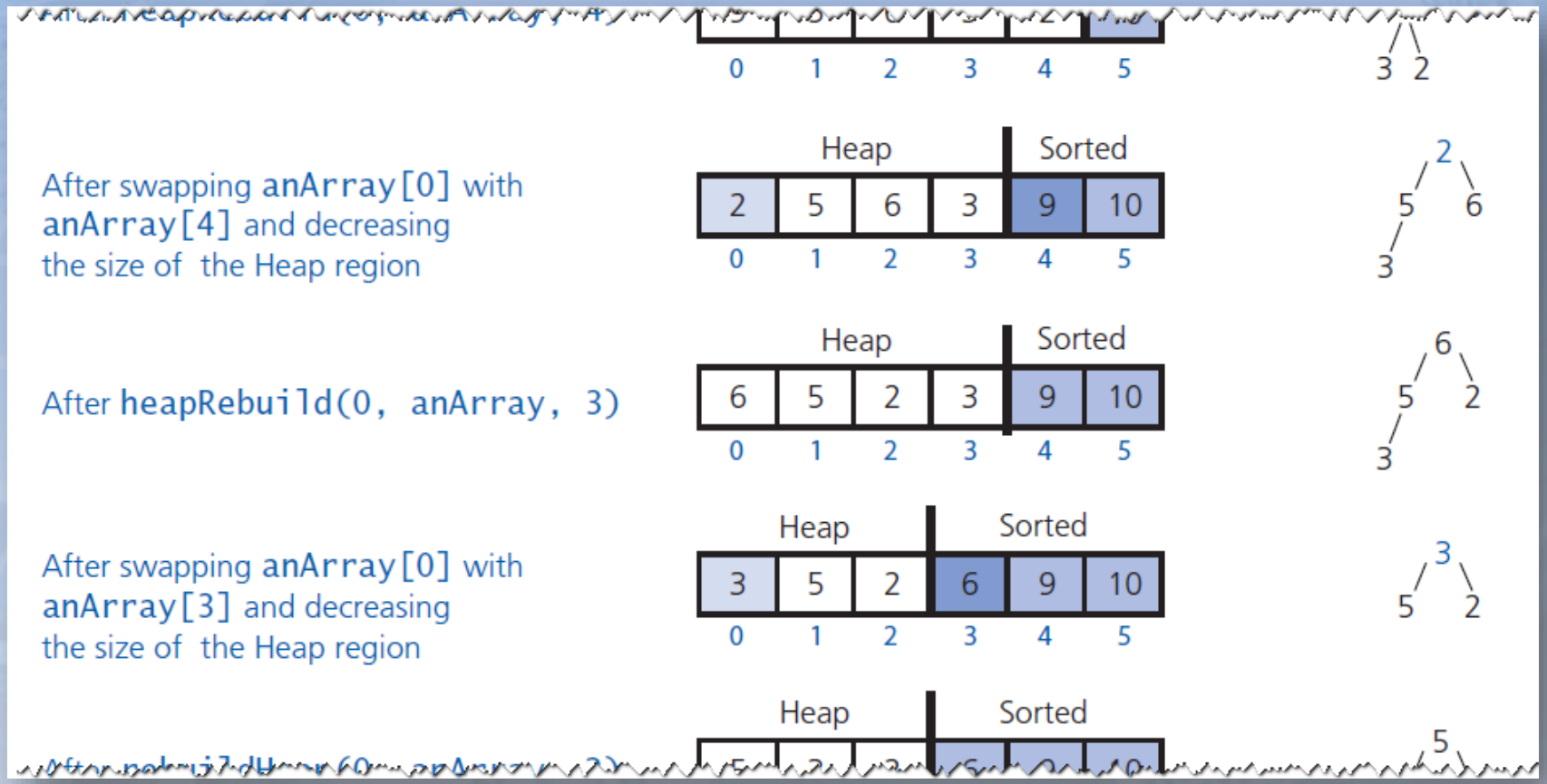

FIGURE 17-11 A trace of heap sort, beginning with the heap in Figure 17-9

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Heap Sort

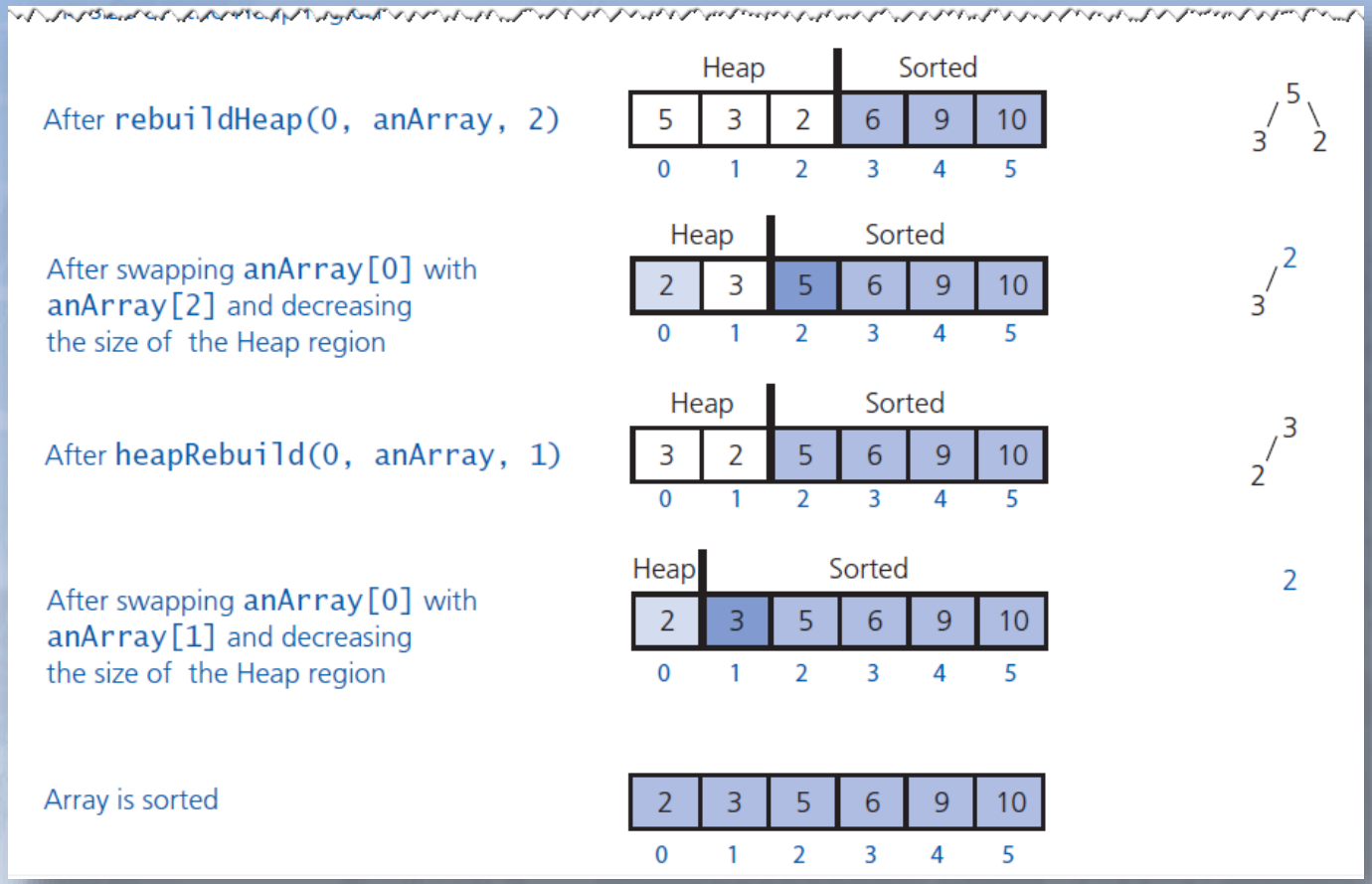

FIGURE 17-11 A trace of heap sort, beginning with the heap in Figure 17-9

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